0363-5465/103/3131-0012$02.00/0
THE AMERICAN JOURNAL OF SPORTS MEDICINE, Vol. 31, No. 1

A Prospective Randomized Study of Patellar versus Hamstring Tendon Autografts for Anterior Cruciate Ligament Reconstruction

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Background: Bone-patellar tendon-bone graft has been the most commonly used graft material in anterior cruciate reconstructions, but there has been increasing use of hamstring tendon grafts. However, no existing clinical studies show adequate support for the choice of one graft over the other.
Hypothesis: Hamstring tendons are equally as good as patellar tendon in anterior cruciate ligament reconstructions.
Study Design: Prospective randomized clinical trial.
Methods: Ninety-nine patients with laxity caused by a torn anterior cruciate ligament underwent arthroscopically assisted reconstruction with graft randomization according to their birth year. Grafts were either bone-patellar tendon-bone with metal interference screw fixation or double-looped hamstring tendons with metal plate fixation. There were no significant differences between the two groups preoperatively or at operation. Standard rehabilitation included immediate postoperative mobilization without a knee brace, protected weightbearing for 2 weeks, and return to full activity at 6 to 12 months.
Results: Forty-three patients in the patellar tendon group and 46 patients in the hamstring tendon group were available for clinical evaluation at a minimum of 21 months after surgery. No statistically significant differences were seen with respect to clinical and instrumented laxity testing, International Knee Documentation Committee Score ratings, isokinetic muscle torque measurements, and Kujala patellofemoral, Lysholm, and Tegner scores.
Conclusion: Equal results were seen for patellar and hamstring tendon autograft anterior cruciate ligament reconstructions at 2 years after surgery. Both techniques seem to improve patients' performance.

Intraarticular ACL reconstructions with autogenous graft material have been widely used for patients with anterior knee laxity, and good-to-excellent results have been reported. The two most commonly used autogenous grafts are the central third of the patellar tendon (bone-patellar tendon-bone) and the hamstring tendon (semitendinosus-gracilis) constructs. Postoperative disadvantages and possible complications of the former include patellofemoral pain, quadriceps muscle weakness, rupture of the patellar tendon, and patellar fracture.[1,12,26] Proposed disadvantages of the latter include failure to achieve immediate rigid fixation to bone[27,30] and lower stiffness compared with a patellar tendon graft or the native ACL.[12,21,30] Despite numerous reports on ACL reconstruction in the literature, we found only four randomized clinical trials comparing patellar and hamstring tendon ACL reconstruction techniques.[2,6,18,22]

The purpose of this study was to determine whether there is a difference in the outcome between two arthroscopically assisted ACL reconstruction techniques: the bone-patellar tendon-bone autograft and the hamstring tendon autograft.

MATERIALS AND METHODS

Patients

Ninety-nine patients were included in this prospective study. The indication for surgery was an ACL rupture confirmed by clinical diagnosis in an otherwise healthy

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No author or related institution has received any financial benefit from research in this study. See "Acknowledgment" for funding information.

patient who experienced instability in activities or wished to maintain his or her preinjury level of activity. The length of time from injury to operation was 19 months (range, 1 week to 20 years) in the patellar tendon group ($N = 51$) and 16 months (range, 2 weeks to 10 years) in the hamstring tendon group ($N = 48$). Originally, 105 patients with ACL instability were included in this prospective study. Two patients in the patellar tendon group and one patient in the hamstring tendon group had had a previous contralateral ACL rupture that was operated on before our randomization protocol started. In addition, one patient in the patellar tendon group and two patients in the hamstring tendon group had an untreated contralateral ACL rupture. These six patients could not be included because all assessments assumed a "normal" contralateral knee. All patients underwent a primary arthroscopically assisted ACL reconstruction and graft randomization into a patellar or hamstring tendon group according to their birth year (even year = patellar tendon, odd year = hamstring tendon). The study was approved by the local ethical committee and the patients gave their informed consent to participate.

Five patients had grade 2 medial collateral ligament (MCL) tears in the ipsilateral knee and four patients had grade 3 MCL tears. Five of these patients were in the patellar tendon group and four were in the hamstring tendon group. Two of the patients with grade 3 tears were treated operatively at the time of the ACL reconstruction (one patient from each group). The rest of the patients with MCL tears were given a knee brace to be worn preoperatively for 6 weeks if the reconstruction was not performed in the acute phase ($N = 2$); otherwise, they received the brace just before or at the time of the operation. All of the MCL tears healed and the patients' knees were stable to medial-lateral stresses on clinical examination throughout the follow-up period.

In 26 patients, 28 meniscal repairs or resections were performed at the time of the ACL reconstruction. In the patellar tendon group there were three partial resections and five repairs of the medial meniscus and six resections and two repairs of the lateral meniscus. In the hamstring tendon group there were four resections and four repairs in the medial meniscus and three resections and one repair in the lateral meniscus. There were fresh osteochondral lesions in eight patients (two in the patellar tendon group and six in the hamstring tendon group). The lesions were from 5×5 mm to 10×10 mm in seven patients; all seven of these lesions were treated with the microfracture technique. In one patient (hamstring tendon group) the lesion was 30×40 mm; it was treated with the chondral cell culture technique immediately after the ACL reconstruction (during the same operation). Second-look arthroscopy performed 1 year later disclosed nearly complete coverage of the lesion.

Operative Technique

Two senior authors (JS and AH) performed all surgeries and both techniques were used equally by each surgeon. All patients were examined under anesthesia (Lachman, drawer, pivot shift, and rotation tests), after which routine diagnostic arthroscopy and necessary meniscal surgery were performed, followed by the ACL reconstruction.

In the patellar tendon group, the surgical procedure was an "outside-in" arthroscopically assisted two-incision technique with the central third of the patellar tendon used as a free autograft with matching drill tunnels made in the femur and tibia (Fig. 1).[4] Specially designed "rear-entry" drill guides were used (Linvatec, Largo, Florida). The fixation of the graft in the drill tunnels was performed with interference metal screws (Linvatec). A 9- or 10-mm drill bit was used and the interference screw was either 8 or 9 mm.

In the hamstring tendon group, the surgical procedure was an arthroscopic single-incision technique with double-looped semitendinosus and gracilis tendons. The drill tunnels were 8 or 9 mm in diameter. Drill guides were used to confirm the correct position of the tunnels. To find the femoral entry point, we used the "bull's eye" drill guide (Linvatec). Proximal graft fixation was achieved with use of a small metal plate (AO, Bern, Switzerland) (Fig. 2).[3,25] The plate was attached to the hamstring tendon graft with a Dacron loop (Davis & Geck, Danbury, Connecticut). The tensile strength of the Dacron loop was 110.8 N, according to the manufacturer. Distal fixation was achieved by tying the graft ends around a cortical 4.5-mm screw secured with a spiked washer (AO).

Postoperative Care

The postoperative care and rehabilitation protocol was the same in both groups. No knee braces were used in postoperative rehabilitation, except when MCL surgery was performed or when a patient with a partially torn MCL was treated with a brace. All knees were immediately mobilized. Full weightbearing was allowed 2 weeks postoperatively. A closed kinetic chain muscle rehabilitation program was employed. Active quadriceps muscle activity was delayed until 3 to 4 weeks postoperatively. Return to sports was allowed gradually, usually without limitations at 6 to 12 months postoperatively.

Follow-up Protocol

Clinical evaluations of knee function and stability were performed preoperatively and 1 and 2 years postoperatively. Lachman and pivot shift testing was performed by the two senior authors (JS and AH). Objective anteroposterior knee stability was determined by using the CA 4000 arthrometer (OSI, Hayward, California), and isokinetic muscle torque of the flexor-extensor system of the knee joint was measured with a dynamometer (Lido Multijoint II, Loredan Biomedical, Inc., West Sacramento, California). Patients underwent routine clinical examination as well as completing the International Knee Documentation Committee (IKDC) evaluation forms.[9] At 1 and 2 years postoperatively, patients completed the Lysholm knee score, Tegner activity level,[31] and Kujala patellofemoral score forms.[16] Anteroposterior and lateral weightbearing radiographs were obtained 2 years postoperatively. Radiographic evaluation was done accord-

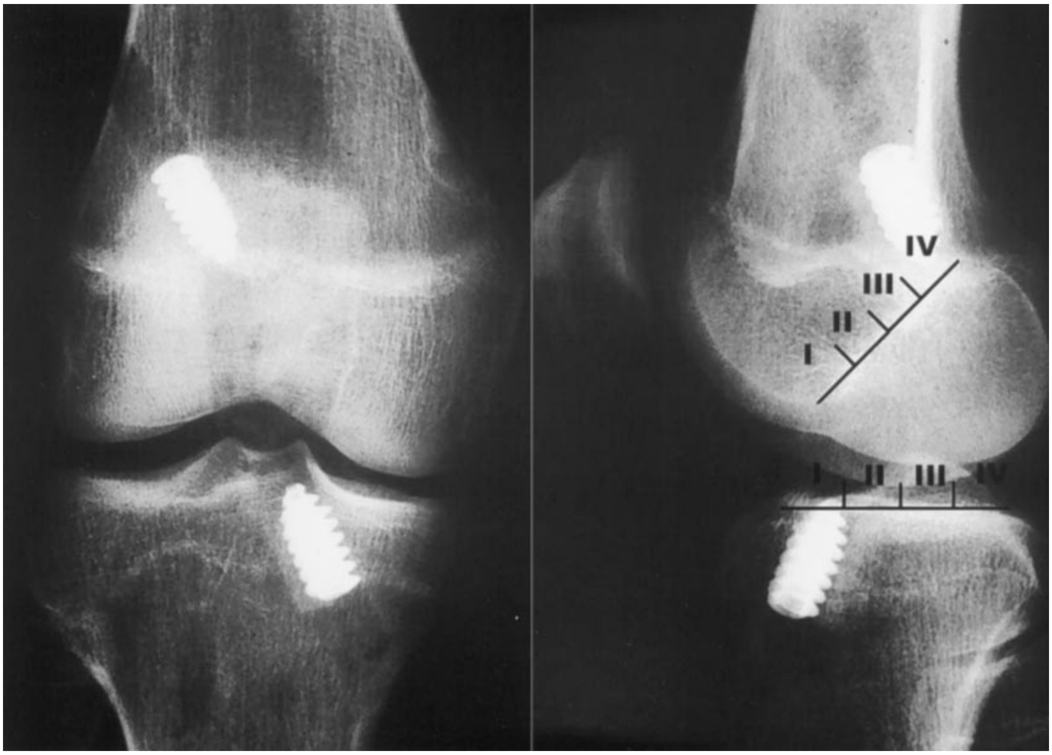

Figure 1. Anteroposterior (left) and lateral (right) radiographs taken 1 day after the operation show the patellar tendon ACL reconstruction technique. The drill tunnel positions in the tibia and femur can be seen on the lateral radiograph. The tibia and femur are divided into four zones.

ing to the IKDC recommendations. Postoperatively, we radiographically evaluated the location of the drill tunnels by dividing the Blumensaat line in the femur and proximal end of the tibia into four zones (I through IV), with zone I being the most anterior (Fig. 1).[8]

Statistical Analysis

Statistical analysis was performed with the BMDP statistical package (Statistical Solutions Ltd., Cork, Ireland). The chi-square test, Student's *t*-test, analysis of variance, and regression analysis were used. The minimum level of significance was $P = 0.05$. The Lysholm score was one of our main outcome measurements. With a difference of 10 points, a clinically significant difference between the treatment groups would be acknowledged. Thus, by accepting less than a 5% probability of a type I error and a power of 80%, a sufficient total sample size would be 54. Having collected a total of 99 cases, we believe that the statistical power was adequate.

RESULTS

Of the 99 total patients in the study, 89 (89.9%) were available for the 2-year follow-up. Of the 10 patients not available, 8 patients (all from the patellar tendon group) were not able to attend the follow-up examination. Follow-up data for the remaining two patients (both in the hamstring tendon group) were excluded because they had ruptured their grafts in an instance of high-energy trauma. These two patients underwent reoperation during the 2-year follow-up period. Therefore, their 2-year evaluation would not represent that of the originally randomized treatment. Thus, 10% of patients were lost to or excluded from follow-up. A total of 89 patients, 43 patients in the patellar tendon group and 46 in the hamstring tendon group, was evaluated at a minimum of 21 months (range, 21 to 38) after surgery.

Preoperative Evaluation

There were no statistically significant differences between the preoperative data of the two groups with respect to sex, age, time from injury to operation, knee stability tests, Tegner activity levels, Kujala patellofemoral scores, or isokinetic muscle torque values. However, in the patellar tendon group, the mean Lysholm score was slightly higher than in the hamstring tendon group (74 versus 68, $P = 0.044$).

Clinical Data and Instrumented Testing

There was a marginal difference between the groups in the range of motion at the 2-year follow-up examination. The mean extension was 0.5° in the patellar tendon group and −0.2° in the hamstring tendon group ($P = 0.048$). There was no difference in flexion, with a mean of 140° of flexion in both groups.

Figure 2. Anteroposterior (left) and lateral (right) radiographs taken 1 day after the operation show the hamstring tendon ACL reconstruction technique.

On clinical examination at the 2-year follow-up there was a positive pivot shift test in three patients in the patellar tendon group and in three patients in the hamstring tendon group (Table 1). The mean time from injury to operation was 6 years in the six patients with a positive pivot shift test 2 years postoperatively, whereas those patients with a negative pivot shift test underwent surgery at a mean of 1 year after injury. The side-to-side difference in anteroposterior knee laxity was 1.7 mm in the patellar tendon group and 1.2 mm in the hamstring tendon group (Table 1). The side-to-side difference increased from 1 year to 2 years by 1.27 mm in the patellar tendon group and by 0.08 mm in the hamstring tendon group (NS, $P = 0.091$). There were no statistically significant differences in the clinical data and instrumented testing results between women and men.

TABLE 1
Results of Clinical and Instrumented Testing 2 Years Postoperatively

Variable	Patellar tendon group	Hamstring tendon group
Pivot shift test (positive/negative)	3/40	3/43
Lachman test (positive/negative)	8/35	8/38
Anteroposterior knee stability, side-to-side difference, mean and range of values (mm)	1.7 (−3.7–7.8)	1.2 (−4.3–7.4)

IKDC Scores

At the 2-year follow-up, there were no statistically significant differences between the groups with respect to overall IKDC scores (Table 2).

Muscle Strength

At the 1-year follow-up, the isokinetic quadriceps muscle torque (percentage of the contralateral knee) at 60 deg/sec was significantly higher in the hamstring tendon group than in the patellar tendon group (85% versus 79%, $P = 0.045$). At the 2-year follow-up, there were no statistically significant differences between the groups with respect to isokinetic muscle performance.

A statistically significant positive correlation ($P < 0.05$) was found between isokinetic extension and flexion torques (60 deg/sec and 180 deg/sec) and Lysholm knee scores at 1 year postoperatively, but this correlation had disappeared by the 2-year follow-up. There was, however, no correlation between isokinetic torque and laxity measurements, placement of the graft as seen on the radiographs, or in the grade of arthrosis.

Lysholm Knee Score

At the 2-year follow-up there were no statistically significant differences between the groups with respect to

TABLE 2
Overall IKDC Scores 2 Years Postoperatively

Rating	Patellar tendon group (N = 43)		Hamstring tendon group (N = 45)[a]	
	N	%	N	%
A (normal)	14	33	18	40
B (nearly normal)	20	46	20	44
C (abnormal)	9	21	7	16
D (severely abnormal)	0	0	0	0

[a] The IKDC score sheets were missing for one patient.

Lysholm knee scores (Table 3). However, the increase in the Lysholm knee score from the preoperative level to that at the 2-year follow-up was significantly greater in the hamstring tendon group (23 points) than in the patellar tendon group (15 points) (P = 0.022).

Kujala Patellofemoral Score

At the 2-year follow-up there were no statistically significant differences between the groups with respect to the Kujala patellofemoral score.

Tegner Activity Level

The 2-year postoperative mean Tegner activity level was 6.1 (range, 2 to 9) in the patellar tendon group and 6.0 (range, 0 to 10) in the hamstring tendon group. There were no statistically significant differences between the groups with respect to Tegner activity level.

Radiographic Evaluation

The postoperative radiographs (N = 96; radiographs from three patients were missing) showed that the drill tunnels were placed in the anterior part of the tibia more often in the patellar tendon group than in the hamstring tendon group. In the patellar tendon group, 14 of 50 tibial tunnels were in zone I, whereas in the hamstring tendon group 3 of 46 were located in zone I (P = 0.0059). There were no differences in the clinical outcome with respect to the placement of drill tunnels.

Analysis of 12 hamstring tendon group cases disclosed that on tibial and femoral AP radiographs, the diameter of the bone tunnel increased during the 2 years of follow-up: from 8.9 mm for the tibial tunnel immediately after operation to 11.7 mm at follow-up (24%, P = 0.003) and from 8.9 mm for the femoral tunnel immediately after operation to 13.2 mm at follow-up (33%, P = 0.01).[11]

TABLE 3
Lysholm Knee Scores 2 Years Postoperatively

Score (points)	Patellar tendon group (N = 43)		Hamstring tendon group (N = 45)	
	N	%	N	%
Excellent (95–100)	22	51	20	45
Good (84–94)	14	33	19	42
Fair (65–83)	6	14	4	9
Poor (<65)	1	2	2	4

Complications

In the patellar tendon group there was one postoperative infection 1 week after the ACL reconstruction. The bacterial culture was negative, but the clinical presentation indicated a purulent arthritis. Arthroscopic debridement and profuse lavage was performed and the patient recovered uneventfully. The knee remained stable and there were no arthritic changes at 2 years postoperatively. There were no deep vein thromboses in either group. Nearly all patients experienced numbness lateral to the tibial incision scar, but there were no signs of a neuroma complication. In the hamstring tendon group, the tibial AO fixation post was removed by 2 years postoperatively in 32 cases in patients who had discomfort at the fixation site.

DISCUSSION

Reports of the metal plate arthroscopic fixation technique in ACL and PCL reconstruction have been published.[3,6,10,11,14,15,17,20,25,28] However, the metal plate fixation ACL reconstruction technique has been used in only a few follow-up studies,[6,10,11,17,20] and comparisons between patellar tendon and hamstring tendon autografts have been made in only three studies.[6,11,17]

L'Insalata et al.[17] retrospectively compared ACL reconstruction with patellar tendon autograft and quadrupled or doubled hamstring tendon autograft. Their graft fixation technique was identical to ours, although they performed tibial fixation in the hamstring tendon group with a bicortical screw and washer. They did not compare the clinical outcome between the groups, but in follow-up radiographs there was enlargement of the bone tunnels after the metal plate fixation method. We have reported the same radiologic phenomenon of bone tunnel enlargement in patients with a hamstring tendon autograft, but there was no correlation with instability.[11] Among reported complications with the metal plate fixation technique are detachment of the plate with migration to the popliteal space[19] and having the fixation plate flip outside the extensor mechanism or vastus lateralis muscle rather than flipping directly outside the lateral femoral cortex.[29]

We found only four randomized clinical trials comparing patellar and hamstring tendon ACL reconstruction techniques.[2,6,18,22] O'Neill[22] evaluated 125 arthroscopically assisted ACL reconstructions after a mean follow-up of 42 months (range, 2 to 5 years). Patients were randomized into three groups according to their month of birth: two-incision reconstruction with use of an autogenous patellar or hamstring tendon graft and single-incision reconstruction (endoscopic technique) with the use of patellar tendon graft. In contrast to our methods, doubled rather than quadrupled hamstring tendon grafts were used and barbed staples were used to fix the autograft. The operative technique in the patellar tendon group was similar to that in our study. O'Neill found no significant differences in the overall outcome between the three groups.

Aglietti et al.[2] performed 63 ACL reconstructions alternating autogenous patellar tendon or doubled (four strands) hamstring tendon grafts. In the patellar tendon group they used outside-in press fit fixation on the femoral side and tied four nonabsorbable sutures over a cortical screw and washer. The tibial side was fixed with an interference screw and four nonabsorbable sutures tied over a cortical screw and washer. In the hamstring tendon group, the femoral side was fixed with a cortical screw and spiked plastic soft tissue washer. The tibial side was fixed with sutures over a cortical screw and washer, and, in 80% of cases (24 of 30), a spiked staple was added. After a mean follow-up of 5.7 years (range, 5.2 to 6.2), 60 patients returned for evaluation. The authors found that objective stability was better in the patellar tendon group than in the hamstring tendon group. Therefore, they recommended the patellar tendon graft for athletes with more serious joint laxity, but they preferred the doubled hamstring tendon autograft in selected cases, such as older patients with a lower functional demand, patients with preexisting patellofemoral problems, and those with failed patellar tendon grafts.

Eriksson et al.[6] recently published a randomized, prospective study conducted in two centers. In this study, 164 patients were randomized to patellar and hamstring tendon groups. The operative technique in the patellar tendon group was similar to that used in our study. However, they used quadrupled semitendinosus tendon with the metal plate fixation technique proximally and double screw post fixation distally. At a median of 31 months (range, 24 to 59) of follow-up, the authors concluded that patellar tendon and quadrupled semitendinosus tendon graft reconstructions have similar outcomes in the medium-term follow-up.

Marder et al.[18] evaluated 72 ACL reconstructions at a mean follow-up of 29 months (range, 24 to 42). Reconstructions were performed on a one-to-one alternating basis with autogenous patellar tendon or doubled hamstring tendon graft. For both graft types the tibial and femoral ends were fixed by sutures over a post and washer. The study demonstrated comparable results in both groups, but there was a statistically significant weakness in peak knee flexion torque at 60 deg/sec in the hamstring tendon group.

In addition to these randomized clinical trials, three prospective nonrandomized studies of arthroscopically assisted ACL reconstruction with autogenous patellar or hamstring tendon grafts have been published.[5,7,23] Results by Corry et al.[5] revealed no differences between the groups in terms of ligament stability, although female patients in the hamstring tendon group showed a slightly increased knee laxity. In contrast, kneeling pain was significantly less common in the hamstring tendon group. Feagin et al.,[7] as well as Otero and Hutcheson,[23] reported that both grafts are good choices in ACL reconstruction, but the patellar tendon graft provided more overall knee stability than did the hamstring tendon graft.

In our present study, the increase in the Lysholm score from preoperatively to the 2-year follow-up was significantly greater in the hamstring tendon group than in the patellar tendon group. This can be explained by the fact that patients in the hamstring tendon group had a lower score preoperatively and they were able to catch up the difference by 2 years postoperatively.

There was a marginal difference in knee extension ability in favor of the hamstring tendon group at the 2-year follow-up examination. This is a logical finding, because the patellar tendon graft is stiffer than the hamstring tendon graft.[12,21,30] A contributing factor might also be that the patellar tendon grafts tended to be placed a little more anteriorly than the hamstring tendon grafts.

The mean time from injury to operation was 6 years in the six patients with a positive pivot shift test 2 years postoperatively, whereas those patients with a negative pivot shift underwent operation at a mean of 1 year after injury. Therefore, it seems that stabilization of the ACL-deficient knee should be done without unnecessary delay, a recommendation also supported by Eriksson et al.[6]

A logical finding was that there was a statistically significant correlation of a good isokinetic extension and flexion torque with a good Lysholm knee score. Also, the extension strength output was better in the hamstring tendon group, as expected at 1 year postoperatively, but at 2 years postoperatively, there was not any difference in the measurements. There was, however, no correlation between isokinetic muscle torque and laxity measurements, placement of the graft as seen on the radiographs, or the grade of arthrosis. In addition, there was no difference in hamstring muscle torque between the operated and control extremity at 2-year follow-up in our study. The reason might be that there is a remarkable regeneration tendency of these tendons, as has been recently shown in MRI studies by Rispoli et al.[24]

In the patellar tendon group, nearly all patients experienced numbness lateral to the tibial incision scar. According to Kartus et al.,[13] there is great variation in the course of the infrapatellar branch of the saphenous nerve. Despite knowledge of these variations, it seems impossible to avoid sectioning these branches and thereby avoid causing sensory deficits.

A limitation in the design of our study is that there were MCL and meniscal injuries in addition to the ACL tear. It is difficult to obtain patients with solitary ACL tears without any other intraarticular lesions. It seems that the additional injuries did not affect the final result and that the injuries were evenly distributed between the two groups. The other limitation of our study is that we did not, in fact, compare patellar and hamstring tendon grafts alone, because the fixation techniques were different. We have preferred using a two-incision technique in the bone-patellar tendon-bone reconstruction so as to avoid impingement of the graft against the screw (if done from the inside), but the hamstring tendon reconstruction with a metal plate fixation is suitable and originally designed for a single-incision technique.

In conclusion, the results of this study demonstrated no clear advantage of either technique at 2 years after surgery. Both techniques seem to improve patients' performance and are, therefore, good choices for ACL reconstruction.

ACKNOWLEDGMENT

This study was supported in part by the Foundation of Orthopedics and Traumatology, Helsinki, Finland.

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